

Grading Policies and College Major Choice with Ability Learning

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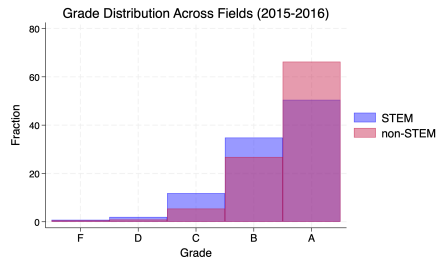
*Disclaimer: The conclusions of this research do not necessarily reflect the opinions or official position of the Texas Education Agency, the Texas Higher Education Coordinating Board, the Texas Workforce Commission or the State of Texas.

Motivation

- College majors are a determining factor of lifetime earnings (Altonji et al., 2014, Webber, 2013).
- What do students consider when choosing courses and majors in college?
 - Interest in the field, comparative advantage, labor market outcomes, on-time graduation
(Arcidiacono, 2004, Wiswall and Zafar, 2015, Zafar, 2013, Andrews et al., 2024, Hsu, 2018)
 - Value a high GPA (Butcher et al., 2014, Ahn et al., 2024)
 - Exploring a field (Stinebrickner and Stinebrickner, 2012, Bordon and Fu, 2015, Arcidiacono et al., 2025)

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 - Exploring a field (Stinebrickner and Stinebrickner, 2012, Bordon and Fu, 2015, Arcidiacono et al., 2025)
- The latter two mechanisms interact directly with departments' grading policies, which experienced significant and uneven grade inflation across fields.



Research Questions

- **Q:** How does **uneven grade inflation** in college across fields affect educational choices?
 - (e.g.) Course choice, major choice, and dropout decision.
- **Q:** How much does it affect **sorting** into majors based on **comparative advantage**?
- **Q:** How much does it affect **gender gap** in major choice?

What This Paper Does

- Build a structural individual decision model of college course and major choices.
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- Model is matched to the [administrative education records](#) of UT-Austin.
 - History of course taking and grades $\Rightarrow$ novel modeling of latent ability and course grades.
 - Identify the taste for a high GPA using a novel policy variation in grade concealing.

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- Model is matched to the [administrative education records](#) of UT-Austin.
 - History of course taking and grades $\Rightarrow$ novel modeling of latent ability and course grades.
 - Identify the taste for a high GPA using a novel policy variation in grade concealing.
- Simulate counterfactual grading policies and full-information benchmark:
 - (1) Equalized to non-STEM level $\Rightarrow$ Uniform inflation
 - (2) Equalized to Avg. B. $\Rightarrow$ Uniform deflation

Contribution to the Literature

- **College major choice:** Arcidiacono, 2004, Wiswall and Zafar, 2015, Wiswall and Zafar, 2018, Wiswall and Zafar, 2021, Zafar, 2013, Bordon and Fu, 2015, Lovenheim and Smith, 2023, Andrews et al., 2024
- **How grades affect educational choices and outcomes:** Bar et al., 2009, Butcher et al., 2014, Bleemer and Mehta, 2024, Denning et al., 2022, Jiang et al., 2021, Ahn et al., 2024, Denning et al., 2025
- **Uncertainty and learning:** Arcidiacono, 2004, Zafar, 2011, Stinebrickner and Stinebrickner, 2012, Bordon and Fu, 2015, Arcidiacono et al., 2025, Larroucau and Rios, 2020, Crawford and Shum, 2005, Hsu, 2018, Papageorgiou, 2014, Guvenen et al., 2020, Bunting et al., 2024
- **Contribution**
 - Implication of **grade inflation** in college on **major sorting** by comparative advantage.
 - **Novel modeling** of latent ability and course grades

Model: Overview

- Majors (m) are grouped into STEM ($m = 1$) and non-STEM ($m = 2$) (Andrews et al., 2024).
- Students have **major-specific** unobserved ability and **update** beliefs about the ability.
 - **Initial beliefs** = population ability distribution (function of gender, parental education, SAT scores).
- Allow preferences to be **gender specific** $\Rightarrow$ What explains gender gap in major choice?

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- Individual decision model with 8 periods:
 - Students choose **a course bundle or dropout** in periods 1-6 and commit to a **major to graduate with** in the 7th period. Graduation payoff and time to graduation realized in the last period.

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- Individual decision model with 8 periods:
 - Students choose **a course bundle or dropout** in periods 1-6 and commit to a **major to graduate with** in the 7th period. Graduation payoff and time to graduation realized in the last period.
- Tradeoffs:
 - **Preference for a high GPA** $\Rightarrow$ Explore less
 - **Match effect between ability and major** $\Rightarrow$ Explore more
 - **Time to graduation** is determined by course taking history. Costly to stay longer in college.

Model: Overview

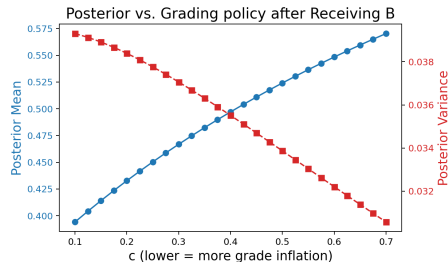
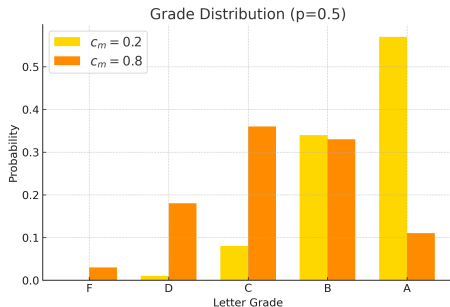
Grading Policy and Belief Updating

- Gaussian models can underestimate the noise of top-coded grades. ▶ Gaussian
- **Course Grades:** $G_{imc} \sim \text{Binomial}(N = 4, a_{im}^{c_m})$ ▶ Mapping
 - $a_{im} \in [0, 1]$: ability to learn in major m .
 - $c_m < 1$: grading policy parameter for field m (low $c_m \Rightarrow$ more grade inflation). ▶ c_m ▶ Mechanism

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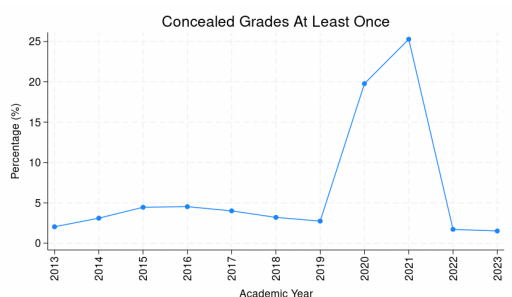
Data: Texas Education Research Center Data

- Administrative educational records of all students who enroll in public institutions in Texas.
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- Complete course taking & grades history linked to major, graduation outcomes, SAT scores, and demographics.
- **Sample:** UT-Austin 2015-2016 Cohorts first-time full-time undergraduate students
 - Summary Statistics
 - Grade inflation is widespread, especially at more selective universities (Rojstaczer and Healy, 2012).
 - Harvard grade inflation report: 79% A in 2020-2021.
- Data facts
 1. GPA residuals are positively correlated with staying in the major. ▸ GPA residuals
 2. Isolating the preference for a high GPA: student's course choices react to grading policies.

Data: Isolating the Preference for a High GPA



	Commitment	Course limit
Pre-pandemic	Before grades	Two electives
Spring 2020	After grades	Unlimited
Fall 2020 – Spring 2021	After grades	Total three

Table: UT-Austin Pass/Fail Grading Policy

- **Results** $\Rightarrow$ Targeted in the model via indirect inference
 1. $\Pr(\text{STEM} > 0)$ increased 1.50-2.13%.
 2. Enrollment share of STEM/Total increased 1.1-1.3%.
 3. Female (3.67%, 1.4%) and Non-STEM (5.56%, 2.8%) more responsive.

► Specification

Estimation

- **Sample:** 2015-2016 cohort to avoid curriculum change and COVID.
- **Two-Stage Estimation:**
 - **First Stage:** Ability distribution parameters and grading policy parameters in each major
 - Identification: Within and across variation in course grades.
 - Estimation Method: Simulated Maximum Likelihood
 - **Second Stage:** Graduation payoffs, time-to-graduation, enrollment costs, etc.
 - Identification: Revealed preference through individuals' choice, conditional on posterior beliefs.
 - Estimation Method: Indirect Inference

► Likelihood ► Indirect Inference ► Model Fit

Graduation Payoff: Value of GPA and the Cost of Extra Semester

	Male	Female
Marginal cost of extra semester to graduate (C)	1.70 (0.048)	2.60 (0.012)
Marginal Value of GPA (β_1)	0.78 (0.019)	1.81 (0.024)
in number of extra semesters (β_1/C)	0.49	0.69

Notes: Standard errors in parentheses.

- Female students dislike extra semester 53% more and value GPA 132% more than male students.

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- Context for interpretation
 - Per semester foregone earnings of college graduates in early career: \$15,000 - \$25,000.
 - Per semester average tuition of in-state public university is around \$5,000.

Counterfactual Grading Policies and Educational Outcomes

	Male			Female		
	Status Quo	Uniform inflation	Uniform deflation	Status Quo	Uniform inflation	Uniform deflation
GPA	3.36	3.49	2.90	3.47	3.56	3.01
Major = STEM (%)	72.68	74.18	70.64	42.16	46.22	41.47
Dropout (%)	6.10	5.51	9.24	2.99	2.34	7.62
Time to graduation	1.96	1.97	1.96	1.67	1.71	1.68

- Equalizing grading policies to non-STEM:

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- Equalizing grading policies to non-STEM:
 - increases **STEM major** choice (1.5-4pp or 2-10%);
 - reduces **dropout** rates (0.6pp or 10-20%);
 - reduces **gender gap** in STEM by 2.6pp (8%).

Grade Inflation $\Rightarrow$ Less Sorting

Figure: Dropout

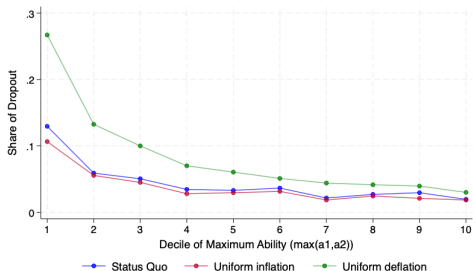
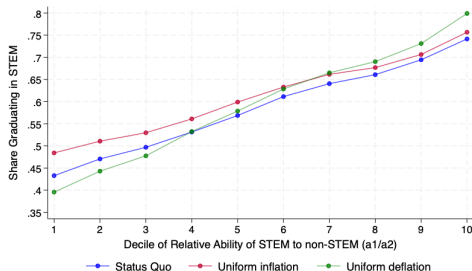


Figure: Major Choice



- 14% less sorting on the **dropout** margin, 9% less sorting on the **major choice** margin

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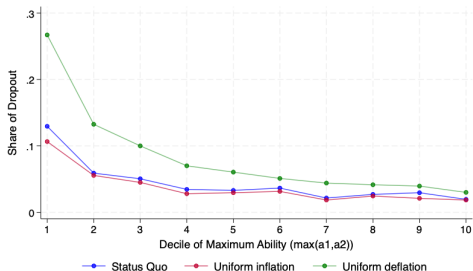
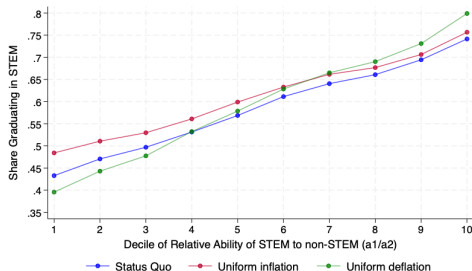


Figure: Major Choice



- 14% less sorting on the **dropout** margin, 9% less sorting on the **major choice** margin
- Two channels:
 1. Reduced incentive to sort based on comparative advantage (high grades everywhere). ► Mechanism
 2. More uncertain about comparative advantage due to noisier signals. ► Mechanism

Conclusions

- Two important channels in college course & major choices that interact with grade inflation.
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- Equalizing STEM grading policy to non-STEM level:
 - ↑ STEM graduation (10%), ↓ dropout rates (20%), ↓ gender gap (8%).
 - Causes less sorting by comparative advantage.
 - 14% less sorting by absolute ability into degree completion.
 - 9% less sorting by comparative ability into STEM major.

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- Two important channels in college course & major choices that interact with grade inflation.
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 - ↑ STEM graduation (10%), ↓ dropout rates (20%), ↓ gender gap (8%).
 - Causes less sorting by comparative advantage.
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 - 9% less sorting by comparative ability into STEM major.
- Grade inflation helps students on observables: degree completion and STEM graduation ↑↑.
- Grade inflation undermines information acquisition and decrease sorting by comparative advantage, likely hurting students in the labor market and lowering aggregate efficiency.

Appendix: Summary Statistics of Pass/Fail Expansion Analysis

Table: Summary Statistics

	Male	Female		Male	Female
Math SAT	1.205 (1.041)	0.572 (1.031)	Graduation rate	85.2%	90.0%
Math SAT missing	9.4 %	8.80%	Initial major is STEM	65.5%	43.5%
Reading SAT	1.037 (1.030)	0.835 (1.027)	Final major is STEM	60.6%	37.3%
Reading SAT missing	9.0%	6.9%	non-STEM STEM	5.6%	14.6%
GPA	3.273 (0.486)	3.402 (0.425)	STEM non-STEM	20.2%	7.0%
GPA in STEM	3.127 (0.626)	3.153 (0.639)	Dropout STEM	19.0%	11.0%
GPA in non-STEM	3.435 (0.506)	3.557 (0.414)	Dropout non-STEM	12.6%	8.7%
Credit hours taken	91.817 (20.857)	93.671 (17.959)			
Share of STEM taken	57.4%	40.1%	N	18,171	21,890

Notes: Standard deviations are in parentheses.

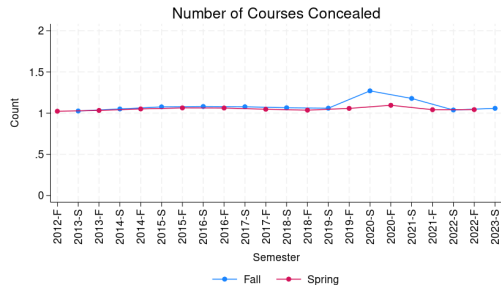
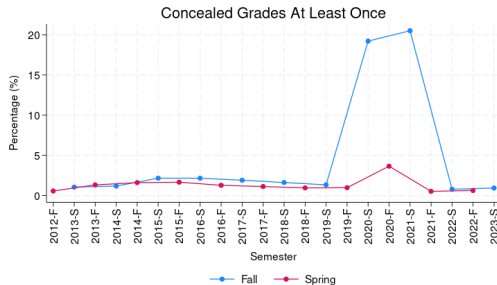
Data: GPA residuals positively correlate with staying in the major

$$GPA_{imt} = \underbrace{\alpha_i}_{\text{Individual FE}} + \underbrace{\eta_{a(it)}}_{\text{Seniority FE}} + \underbrace{\rho_{q(imt)}}_{\text{Course portfolio FE}} + \varepsilon_{imt} \quad \text{for } m = 1, 2 \quad (1)$$

$$\underbrace{STEM_i}_{\substack{1 \text{ if major is } m = 1 \\ \text{in semester 7}}} = \underbrace{\theta_1}_{\substack{0.2165 \\ (0.0024)}} \underbrace{\sum_t \left(\frac{\hat{\varepsilon}_{i1t}}{6} \right)}_{\substack{\text{Residual grade} \\ \text{average} \\ \text{in STEM}}} + \underbrace{\theta_2}_{\substack{-0.1378 \\ (0.0025)}} \underbrace{\sum_t \left(\frac{\hat{\varepsilon}_{i2t}}{6} \right)}_{\substack{\text{Residual grade} \\ \text{average} \\ \text{in non-STEM}}} + X_i' \Gamma + v_{it} \quad (2)$$

- X_i : Math & Reading SAT, initial major, gender, parental education level.
- 1 SD increase in **STEM** grade residual $\Rightarrow$ **31pp increase** in STEM declaration.
- 1 SD increase in **non-STEM** grade residual $\Rightarrow$ **20pp decrease** in STEM declaration.
- Consistent with ability learning through course grades story.

Appendix: Concealing Decision



- Fraction of students concealing at least one grade rose from 1.4% to 19.2% in Spring 2020.
- After a dip in Fall 2020 to 3.7%, it rebounds to 20.5% in Spring 2021.
- Avg. N of courses concealed remained steady with a rise to 1.2 during the flexible pass/fail period.

Appendix: Empirical Strategy

$$Y_{it} = \alpha + \beta_0 \underbrace{\text{Expect}_{it}}_{\text{Fall 2020 dummy}} + \beta_1 \underbrace{\text{PF}_{it}}_{\text{Spring 2021 dummy}} + \underbrace{\delta_{am(it)}}_{\substack{\text{Academic} \\ \text{seniority} \\ \times \text{major FEs}}} + \underbrace{\gamma_i}_{\text{Student FE}} + \underbrace{S'_{it}\Lambda}_{\text{Time-varying controls}} + \varepsilon_{iat}$$

- Exploit time variation in UT-Austin's pass/fail policy and compare students with similar seniority, prior grades, and course loads.
- Time-varying controls include cumulative credit hours taken in STEM and non-STEM, cumulative GPA in STEM and non-STEM, separately.
- Test if a more generous pass/fail policy encouraged risk-taking in course selection (i.e., taking harsher-graded STEM courses).

Data: Isolating the Preference for a High GPA

- Empirical strategy: Compare students across time + Placebo test on other universities.

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- Results (β_0, β_1)
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- **Mapping to the model:** indirect inference by simulating grade concealing in the model.

Appendix: Results - Extensive Margin

	(1)	(2)	(3)	(4)	(5)
Sample	All	Female	Male	non-STEM	STEM
<i>Expect</i>	0.017*** (0.003)	0.020*** (0.005)	0.013*** (0.004)	0.030*** (0.007)	0.007*** (0.002)
<i>PF</i>	0.012*** (0.003)	0.013*** (0.005)	0.010*** (0.004)	0.022*** (0.007)	0.004* (0.002)
Mean	0.797	0.545	0.867	0.539	0.976
SD	0.402	0.441	0.340	0.498	0.154

Notes: Standard errors are clustered at the individual level. * p-value < 0.1, ** p-value < 0.05, and *** p-value < 0.001.

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Appendix: Results - Intensive Margin

	(1)	(2)	(3)	(4)	(5)
Sample	All	Female	Male	non-STEM	STEM
<i>Expect</i>	0.007*** (0.002)	0.006** (0.003)	0.008** (0.003)	0.005 (0.003)	0.007*** (0.003)
<i>PF</i>	0.006*** (0.002)	0.004 (0.003)	0.007** (0.003)	0.006** (0.003)	0.003 (0.003)
Mean	0.523	0.437	0.620	0.213	0.737
SD	0.357	0.349	0.341	0.245	0.250

Notes: Standard errors are clustered at the individual level. * p-value < 0.1, ** p-value < 0.05, and *** p-value < 0.001.

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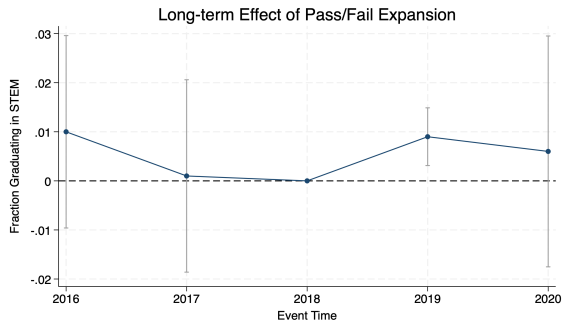
Appendix: Long-term Effect of COVID Pass/Fail Grading Policy

$$Y_i = \sum_{k \neq -1} \beta_k \cdot (\mathbf{1}\{t(i) - 2019 = k\} \cdot D_i) + \alpha_{s(i)} + \lambda_{t(i)} + X_i' \Gamma + \varepsilon_{it} \quad (3)$$

- Y_i : Whether i graduated in STEM in 5 years.
- D_i is a dummy that takes 1 if i 's school continued Pass/Fail expansion after Spring 2020.
- β_k identifies the treatment effect k periods away from the event.
- $\alpha_{s(i)}$: School fixed effect.
- λ_t : Cohort fixed effect.
- X_i : Subject SAT scores, gender, parental education level, initial intended major.

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Appendix: Long-term Effect of COVID Pass/Fail Grading Policy



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Data: Placebo Test

Pass/Fail in Spring 2021

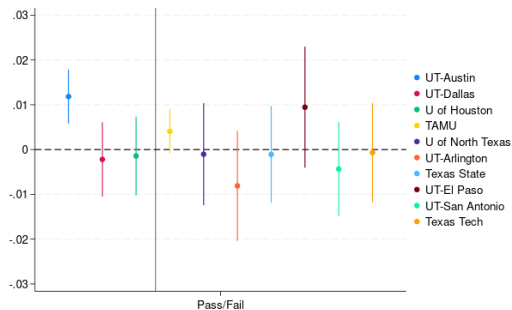


Figure: Probability of Taking Any STEM Course

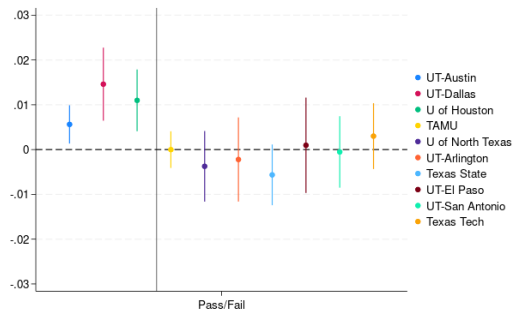


Figure: Share of STEM Courses Enrolled

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Appendix: Placebo Test

Expectation in Fall 2020

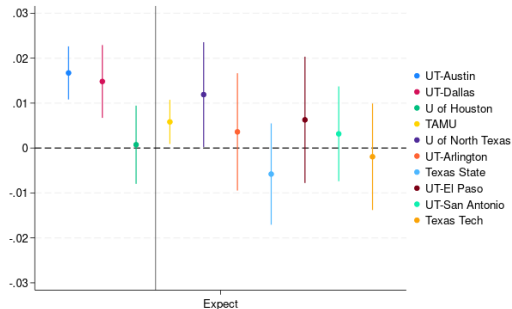


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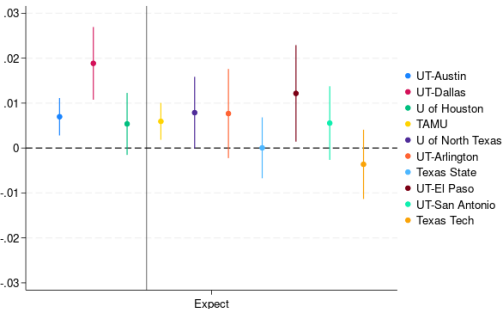
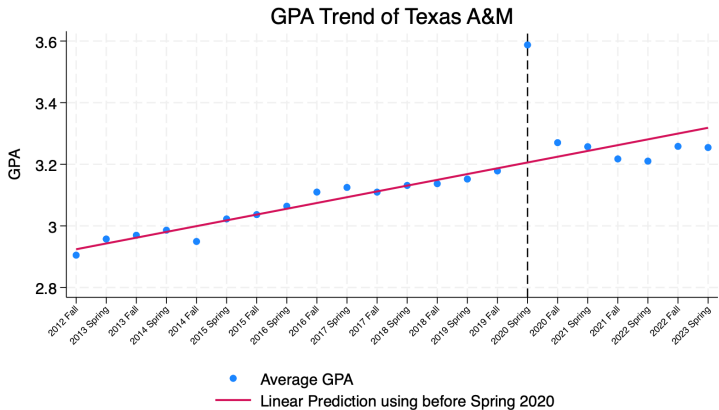


Figure: Share of STEM Courses Taken

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Appendix: GPA Trend of Texas A&M



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Appendix: Model - Optimization Problem

$$\text{State space: } \Omega_{it} = \left\{ \overbrace{X_i}^{\text{Gender}}, \overbrace{\vec{\mu}_{it}}^{\text{Mean of Belief on Ability}}, \overbrace{\vec{\sigma}_{it}}^{\text{Variance of Belief on Ability}}, \overbrace{\vec{Q}_{it}}^{\text{N of courses taken}}, \overbrace{G_{it}}^{\text{GPA in period t}} \right\}$$

Appendix: Model - Optimization Problem

$$\text{State space: } \Omega_{it} = \left\{ \overbrace{X_i}^{\text{Gender}}, \overbrace{\vec{\mu}_{it}}^{\text{Mean of Belief on Ability}}, \overbrace{\vec{\sigma}_{it}}^{\text{Variance of Belief on Ability}}, \overbrace{\vec{Q}_{it}}^{\text{N of courses taken}}, \overbrace{G_{it}}^{\text{GPA in period } t} \right\}$$

$$\max\left\{ \overbrace{\max_{\vec{q}} U_t(\vec{q}, \Omega_{it})}^{\text{Flow cost of taking bundle } \vec{q}} + \overbrace{0.95 \cdot \mathbb{E}_{\Omega'_{i,t+1}} [V_{t+1}(\Omega'_{i,t+1}) | \Omega_{it}, \vec{q}]}^{\text{Continuation value}} + \overbrace{\eta_{i\vec{q}t}}^{\text{Preference shocks}}, \eta_{iDt} \right\} \text{ for } t = 1, \dots, 6$$

$$\max\left\{ \max_{m \in \{1,2\}} \mathbb{E}_{\Omega_{iT}} \left[\underbrace{V_{im}(\Omega_{iT})}_{\text{Graduation payoff}} \right] - \underbrace{C(X_i) \cdot \mathbb{E}_{\Omega_{iT}} [T_{im}]}_{\text{Cost of extra time}} + \underbrace{\xi_{im}}_{\text{Preference shocks}}, \xi_{iD} \right\} \text{ in } T = 7$$

Appendix: Preference Shock Distribution

$$F_{\eta_t}(\eta_{Dt}, \{\eta_{\vec{q}t}\}_{\vec{q} \in Q}) = \exp\left(-e^{-\eta_{Dt}/v} - \left(\sum_{\vec{q} \in Q} e^{-\eta_{\vec{q}t}/(v\lambda_Q)}\right)^{\lambda_Q}\right) \text{ for } t = 1, \dots, 6$$

$$F_{\xi}(\xi_D, \{\xi_m\}_{m \in \{1,2\}}) = \exp\left(-e^{-\xi_D} - \left(\sum_{m \in \{1,2\}} e^{-\xi_m/\lambda_M}\right)^{\lambda_M}\right)$$

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Appendix: Model - Flow Payoff

$$\max\{\underbrace{\max_{\vec{q}} \overbrace{U_t(\vec{q}, \Omega_{it})}^{\text{Flow cost of taking bundle } \vec{q}}} + \underbrace{0.95 \cdot \mathbb{E}_{\Omega'_{i,t+1}} [V_{t+1}(\Omega'_{i,t+1}) | \Omega_{it}, \vec{q}]}_{\text{Continuation value}} + \eta_{i\vec{q}t}\}_{, \eta_{iDt}} \text{ for } t = 1, \dots, 6$$

$$U_t(\vec{q} = (q_1, q_2), \Omega_{it}) = \underbrace{\bar{\gamma}(X_i)}_{\text{Enrollment Cost}} + \underbrace{\gamma_1(X_i)}_{\text{MC of STEM}} \cdot q_1 + \underbrace{\gamma_2(X_i)}_{\text{MC of non-STEM}} \cdot q_2 \quad (4)$$

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Appendix: Model - Graduation Payoff

$$\max\left\{\max_{m \in \{1,2\}} \left\{ \underbrace{\mathbb{E}_{\Omega_{iT}}[V_{im}(\Omega_{iT})]}_{\text{Graduation payoff}} - \underbrace{C(X_i) \cdot \mathbb{E}_{\Omega_{iT}}[T_{im}]}_{\text{Cost of extra time}} + \xi_{im} \right\}, \xi_{iD}\right\}$$

$$V_{im} = \underbrace{\beta_{0m}(X_i)}_{\text{Mean payoff}} + \beta_1(X_i) \underbrace{GPA_i}_{\text{Final GPA}} \quad (5)$$

$$+ \beta_{2m}(X_i) \underbrace{a_{im}}_{\text{Ability in field } m} + \beta_{3m}(X_i) \underbrace{S_{im}}_{\text{Share of course in field } m} + \beta_{4m}(X_i) \underbrace{(a_{im} \cdot S_{im})}_{\text{Match effect}} \quad (6)$$

$$V_{i0} = 0$$

Appendix: Model - Graduation Payoff

$$\max\left\{\max_{m \in \{1,2\}} \underbrace{\{\mathbb{E}_{\Omega_{iT}}[V_{im}(\Omega_{iT})]\}_{}}_{\text{Graduation payoff}} - \underbrace{C(X_i) \cdot \mathbb{E}_{\Omega_{iT}}[T_{im}]}_{\text{Cost of extra time}} + \xi_{im}\right\}, \xi_{iD}$$

$$T_{im}(\Omega_{i7}) \sim \text{Poisson}(\lambda_{im}) \quad (3)$$

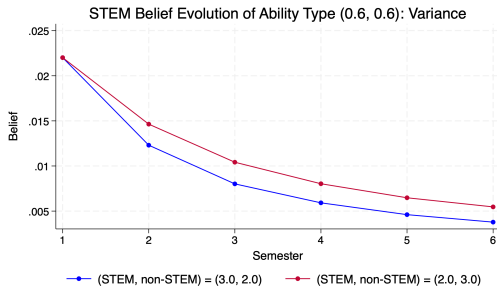
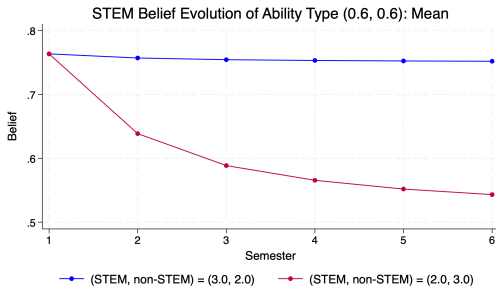
$$\text{s.t. } \log(\lambda_{im}) = \tau_{0m} + \tau_{1m} \underbrace{Q_{i7m}}_{\substack{\text{Cumulative} \\ \text{N of courses} \\ \text{taken in } m}} + \sum_{k \neq m} \tau_{2k} Q_{i7k} + \sum_{k \neq m} \tau_{3k} Q_{i7m} Q_{i7k}$$

► w/ Grade Concealing ► Back to Main

Appendix: Model Mechanism

► Back to Main

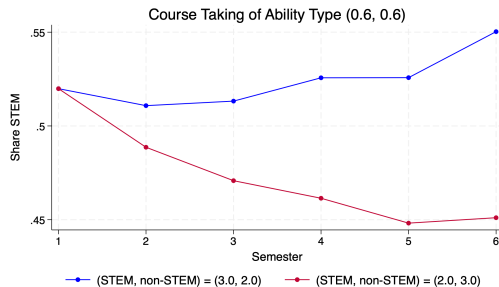
- Fix ability, always give the same grades $\Rightarrow$ How do beliefs and choices evolve?



Appendix: Model Mechanism

► Back to Main

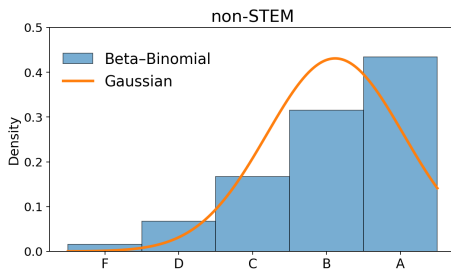
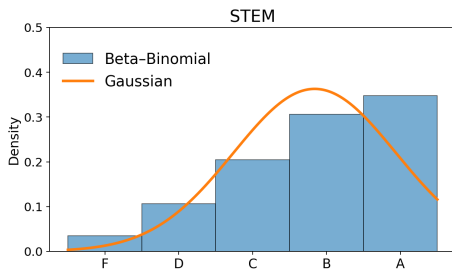
- Fix ability, always give the same grades $\Rightarrow$ How do beliefs and choices evolve?



- Graduation in STEM: 46.50% vs. 36.74%

Appendix: Model - Ability, Beliefs, and Grading Policy

- Gaussian models can underestimate the noise of top-coded grades.
 - Suppose $a_{im} \sim \text{Normal}(\mu_{im}, \sigma_{im}^2)$ and $G_{im} = \bar{g}_m + a_{im} + \varepsilon_{im}$ where $\varepsilon_{im} \sim \text{Normal}(0, v_m^2)$
 - Gaussian predicts non-STEM sends more accurate signal: $v_1 = 1.07 > v_2 = 0.89$



Appendix: Grading Policy

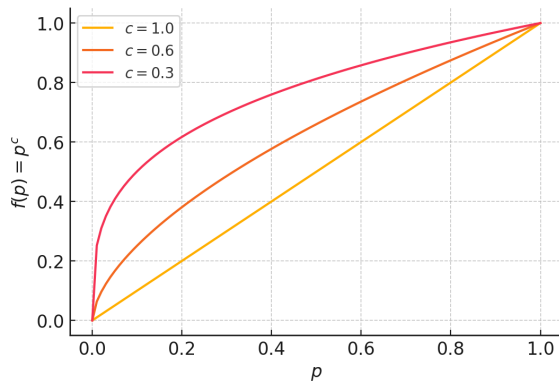


Figure: How c_f Distorts the Relationship Between p_{if} and g_{ifc}

Appendix: Grading Scheme

Table: Mapping of Letter Grade - g_{ifc} - Grade Points

Letter Grade	g_{ifc}	Grade Points
A	4	4.0
B	3	3.0
C	2	2.0
D	1	1.0
F	0	0.0

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Appendix: Belief Updating Rule

- Prior beliefs start from the population distribution of ability.
- Upon receiving a stream of grades $\{g_{imc}\}_{c \in C_m}$, update beliefs by updating α'_{im} and β'_{im} .

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$$\begin{aligned} f_{im}^{Posterior}(a) &\propto \overbrace{a^{\alpha_{im}-1} \times (1-a)^{\beta_{im}-1}}^{\text{Contribution from the prior}} \times \overbrace{\{(a^{c_m})^{g_{imc}} \times (1-a^{c_m})^{N-g_{imc}}\}}^{\text{Contribution from the likelihood}} \\ \mathbb{E}(a_{im}) &= \int a \cdot f_{im}^{Posterior}(a) da, \quad \mathbb{V}(a) = \int (a - \mathbb{E}(a))^2 \cdot f_{im}^{Posterior}(a) dp \end{aligned} \tag{5}$$

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- Updating rule:

$$\alpha'_{im} = \mathbb{E}(a_{im}) \cdot \left(\frac{\mathbb{E}(a_{im}) \cdot (1 - \mathbb{E}(a_{im}))}{\mathbb{V}(a_{im})} - 1 \right), \quad \beta'_{im} = (1 - \mathbb{E}(a_{im})) \cdot \left(\frac{\mathbb{E}(a_{im}) \cdot (1 - \mathbb{E}(a_{im}))}{\mathbb{V}(a_{im})} - 1 \right) \quad (6)$$

Appendix: Estimation - First Stage (Grading Policies & Ability Distributions)

- Θ_a : Ability distribution parameters, grading policy parameters.
- Individual i 's Likelihood Contribution:

$$L_i(\{g_{imkt}\}_{m,k,t}|\Theta_a) = \int L(\{g_{imk1}\}_{m,k}|\Theta_a, A) \times L(\{g_{imk2}\}_{m,k}|\{g_{imk1}\}_{m,k}, \Theta_a, A) \cdots \times L(\{g_{imk6}\}_{m,k}|\{g_{imk1}\}_{m,k}, \cdots, \{g_{imk5}\}_{m,k}, \Theta_a, A) dF_A. \quad (7)$$

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- How can the course grades likelihood be separated from the choice likelihood?
 - Initial beliefs = Population ability distribution (assumption)
 - Initial beliefs + Sequence of course grades $\Rightarrow$ Control selection into courses.

Appendix: Estimation - Second Stage (Preference Parameters)

- Θ_d : Graduation payoffs, time-to-graduation, enrollment costs, cost of grade concealing, etc.
- Indirect inference

$$\hat{\Theta}_d = \arg \min_{\Theta_d \in \Theta} \left(\hat{\beta}^{\text{data}} - \hat{\beta}^{\text{sim}}(\Theta_d) \right)^{\top} W \left(\hat{\beta}^{\text{data}} - \hat{\beta}^{\text{sim}}(\Theta_d) \right) \quad (8)$$

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- Moments matched
 1. Regression coefficients of major choice on GPA, beliefs, and course taking.

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- Moments matched
 1. Regression coefficients of major choice on GPA, beliefs, and course taking.
 2. Regression coefficients of extra time to graduate on number of courses taken in each field.

Appendix: Estimation - Second Stage (Preference Parameters)

- Θ_d : Graduation payoffs, time-to-graduation, **enrollment costs**, cost of grade concealing, etc.
- Indirect inference

$$\hat{\Theta}_d = \arg \min_{\Theta_d \in \Theta} \left(\hat{\beta}^{\text{data}} - \hat{\beta}^{\text{sim}}(\Theta_d) \right)^{\top} W \left(\hat{\beta}^{\text{data}} - \hat{\beta}^{\text{sim}}(\Theta_d) \right) \quad (8)$$

- Moments matched
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 2. Regression coefficients of extra time to graduate on number of courses taken in each field.
 3. Dropout rates, major choice rates, and average extra time to graduate by major.
 4. Average number of courses taken in each field by chosen major.

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- Moments matched

1. Regression coefficients of major choice on GPA, beliefs, and course taking.
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3. Dropout rates, major choice rates, and average extra time to graduate by major.
4. Average number of courses taken in each field by chosen major.
5. $\Delta \Pr(\text{STEM} > 0)$ and $\Delta(\text{Share of STEM courses taken})$ ▶ Grade Concealing
6. Fraction of students who choose to conceal the lowest grade when concealing is available.

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Appendix: Grade concealing in the second stage estimation

- **Assumption:** Each period, conceal the lowest grade when it's profitable.
- Let $PF_{i,t}$ be the number of concealed grades for i at the beginning of t .

$$\max_{\vec{q}} \underbrace{U_t(\vec{q}, \Omega_{it})}_{\text{Flow cost of taking bundle } \vec{q}} + 0.95 \cdot \underbrace{\mathbb{E}_{\Omega'_{i,t+1}|\Omega_{it}}[V_{t+1}(\Omega'_{i,t+1}, PF_{i,t+1}|\Omega_{it}, PF_{it})|\vec{q}]}_{\text{Continuation value}} + \underbrace{\sigma(\eta_{idt} + \lambda^q \eta_{i\vec{q}t})}_{\text{preference shock}} \quad (9)$$

- For a drawn grade shock and thus $\Omega'_{i,t+1}$,

$$V_{t+1}(\Omega'_{i,t+1}, PF_{i,t+1}) = \begin{cases} V_{t+1}(\Omega'_{i,t+1}, PF_{it} + 1) & \text{if } V_{t+1}(\Omega'_{i,t+1}, PF_{it} + 1) > V_{t+1}(\Omega'_{i,t+1}, PF_{it}) \\ V_{t+1}(\Omega'_{i,t+1}, PF_{it}) & \text{otherwise} \end{cases} \quad (10)$$

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- Let $PF_{i,t}$ be the number of concealed grades for i at the beginning of t .

$$\max_{\vec{q}} \underbrace{U_t(\vec{q}, \Omega_{it})}_{\text{Flow cost of taking bundle } \vec{q}} + 0.95 \cdot \underbrace{\mathbb{E}_{\Omega'_{i,t+1}|\Omega_{it}}[V_{t+1}(\Omega'_{i,t+1}, PF_{i,t+1}|\Omega_{it}, PF_{it})|\vec{q}]}_{\text{Continuation value}} + \underbrace{\sigma(\eta_{idt} + \lambda^q \eta_{i\vec{q}t})}_{\text{preference shock}} \quad (9)$$

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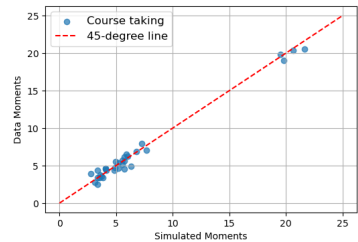
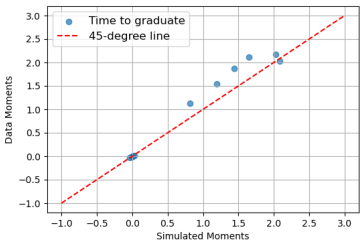
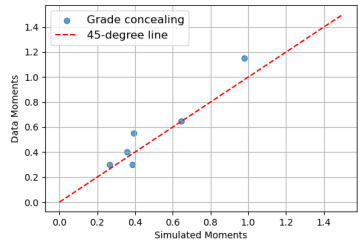
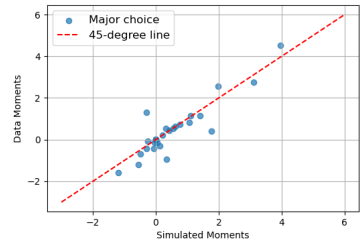
$$\text{where } V_{im} = \beta_{0m} + \beta_1 GPA_i + \beta_{2m} p_{im} + \beta_{3m} S_{im} + \beta_{4m} (p_{im} \cdot S_{im}) - \underbrace{\beta_5 PF_{i7}}_{\text{Cost of Concealing}}, \quad V_0 = 0 \quad (11)$$

Appendix: Estimation Results

Table: Estimates: Flow Utility and Graduation Payoff

	Male		Female	
	STEM	non-STEM	STEM	non-STEM
Flow utility				
Enrollment (γ_0)		-1.67 (0.036)		-1.71 (0.017)
per 1 course (γ_m)	-0.11 (0.023)	-0.07 (0.001)	-0.13 (0.075)	-0.12 (0.001)
Graduation payoff				
Mean utility (β_{0m})	0.77 (0.011)	1.03 (0.029)	0.44 (0.035)	0.94 (0.011)
Taste for GPA (β_1)		0.78 (0.019)		1.81 (0.024)
Ability in my major (β_{2m})	-1.97 (0.012)	-2.03 (0.004)	-2.05 (0.042)	-2.29 (0.040)
Share of courses (β_{3m})	3.57 (0.024)	3.30 (0.031)	3.79 (0.027)	3.22 (0.060)
Ability $\times$ Share of courses (β_{4m})	13.85 (0.013)	7.44 (0.015)	10.72 (0.081)	7.94 (0.032)
Cost of extra semester (C)		1.70 (0.048)		2.60 (0.012)
Cost of concealing 1 grade		0.06 (0.0002)		0.15 (0.005)

Appendix: Model Fit



Appendix: Other Parameter Estimates I

Table: Estimates: Ability Distribution and Grading Policies

	STEM		non-STEM	
	Coeff	SE	Coeff	SE
Ability parameters				
α : mean	0.635	(0.075)	0.682	(0.067)
α : H parental edu.	0.289	(0.038)	0.229	(0.038)
α : female	0.174	(0.046)	0.319	(0.039)
β : mean	1.537	(0.076)	1.315	(0.074)
β : H parental edu.	-0.287	(0.081)	-0.399	(0.073)
β : female	0.446	(0.056)	0.093	(0.044)
SAT: intercept	0.884	(0.032)	0.692	(0.022)
SAT: slope	1.295	(0.057)	0.765	(0.033)
SAT: scale	4.338	(0.012)	4.301	(0.011)
Grading policy				
Year 1	0.166	(0.015)	0.120	(0.007)
Year 2	0.178	(0.016)	0.120	(0.007)
Year 3	0.166	(0.015)	0.103	(0.006)

Appendix: Other Parameter Estimates II

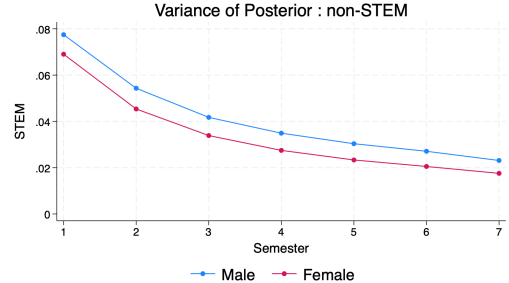
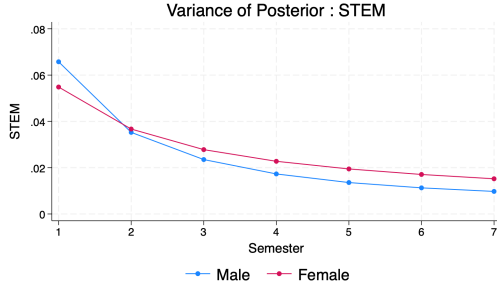
Table: Estimates: Extra Time to Graduation

	STEM		non-STEM	
	Coeff	SE	Coeff	SE
τ_0	0.368	(0.007)	1.490	(0.002)
τ_1	0.017	(0.0002)	-0.066	(0.0006)
τ_2	0.039	(0.0004)	-0.074	(0.0002)
τ_3	-0.002	(0.00001)	0.007	(0.00004)

Table: Estimates: Preference Shock

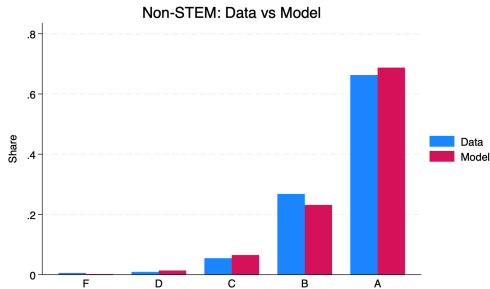
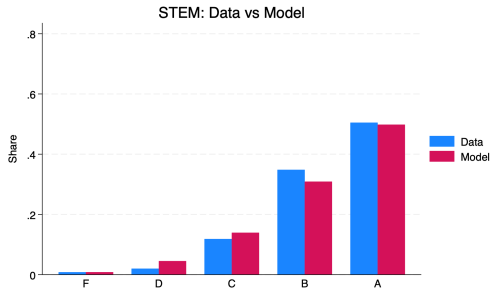
	Coeff	SE
v_1	1.13	(0.006)
v_2	0.87	(0.003)
v_3	0.96	(0.007)
λ_Q	0.77	(0.002)
λ_M	0.29	(0.015)

Appendix: Grading Policies and Learning



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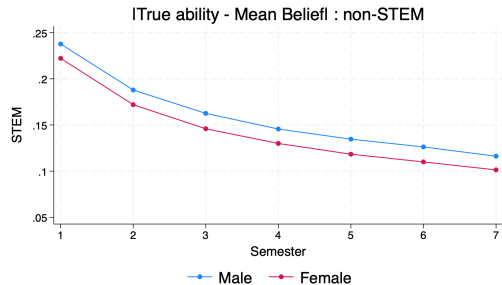
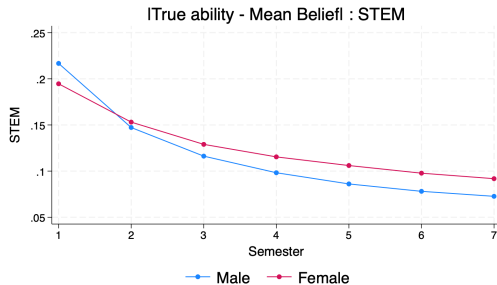
Grading Policies



- Grade distribution is highly left skewed, more top-coding in non-STEM.
 - $c_m = \text{STEM } 0.166$ (s.e. 0.015) vs. non-STEM 0.120 (s.e. 0.007)
 - For an average student to receive B on average: $c \approx 0.30$ (STEM) vs. 0.42 (non-STEM)

Convergence of Beliefs to True Ability

► Variance of belief



- Distance: Absolute value of (True ability - Mean belief)
- Students' belief converge to their true ability but on average 0.3-0.4 SD gap left.
- Female students learn more about non-STEM ability, male students learn better about STEM ability.

Appendix: Grade Inflation $\Rightarrow$ Less Sorting - GPA preference channel

Table: STEM Graduation Rate* (Under Full-information)

Relative Ability (a_1/a_2)	Q1	Q2	Q3	Q4
Uniform inflation	45.67	55.81	66.28	75.56
Uniform deflation	38.73	52.83	66.60	79.25

* Conditional on not dropping out.

- When ability is known, grades matter only because students value GPA (no learning channel).
- When grading is lenient, the cost of taking a difficult or mismatched course falls.

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Appendix: Grade Inflation $\Rightarrow$ Less Sorting - Ability learning channel

Table: STEM Graduation Rate* (With Uncertainty)

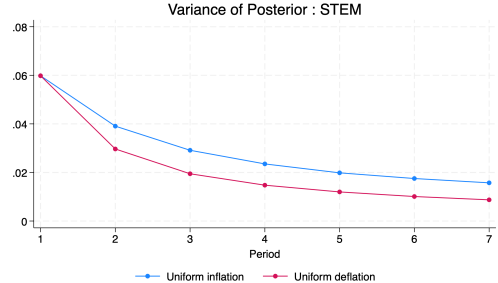
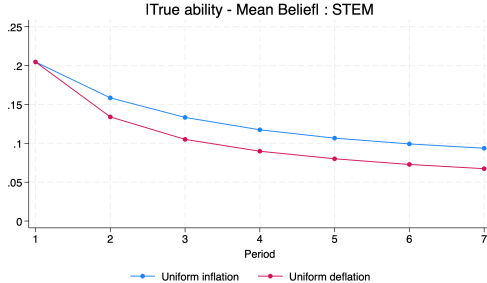
Relative Ability (a_1/a_2)	Q1	Q2	Q3	Q4
Uniform inflation	50.21	57.18	65.05	72.32
Uniform deflation	43.16	54.10	65.23	75.27

* Conditional on not dropping out.

- Students remain uncertain about their fit. More uncertain under uniform inflation.
- Weaker sorting compared to the full information benchmark under both regimes.

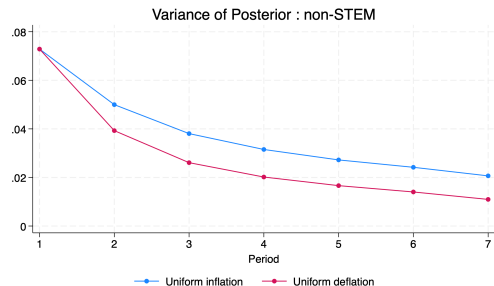
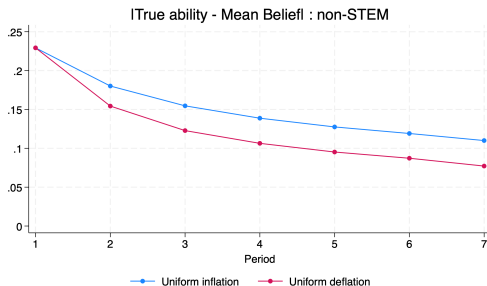
► Learning: STEM ► Learning: non-STEM ► Back to main

Appendix: STEM Ability Learning Under Uncertainty



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Appendix: non-STEM Ability Learning Under Uncertainty



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References I

-  Ahn, Tom et al. (2024). “Equilibrium grading policies with implications for female interest in STEM courses”. In: [Econometrica](#) 92.3, pp. 849–880.
-  Altonji, Joseph G, Lisa B Kahn, and Jamin D Speer (2014). “Trends in earnings differentials across college majors and the changing task composition of jobs”. In: [American Economic Review](#) 104.5, pp. 387–93.
-  Andrews, Rodney J et al. (2024). “The returns to college major choice: Average and distributional effects, career trajectories, and earnings variability”. In: [Review of Economics and Statistics](#), pp. 1–45.
-  Arcidiacono, Peter (2004). “Ability sorting and the returns to college major”. In: [Journal of Econometrics](#) 121.1-2, pp. 343–375.

References II

-  Arcidiacono, Peter et al. (2025). “College attrition and the dynamics of information revelation”. In: [Journal of Political Economy](#) 133.1, pp. 53–110.
-  Bar, Talia, Vrinda Kadiyali, and Asaf Zussman (2009). “Grade information and grade inflation: The Cornell experiment”. In: [Journal of Economic Perspectives](#) 23.3, pp. 93–108.
-  Bleemer, Zachary and Aashish Mehta (2024). “College Major Restrictions and Student Stratification”. In: [Nature](#). Revisions requested.
-  Bordon, Paola and Chao Fu (2015). “College-major choice to college-then-major choice”. In: [The Review of economic studies](#) 82.4, pp. 1247–1288.
-  Bunting, Jackson, Paul Diegert, and Arnaud Maurel (2024). [Heterogeneity, uncertainty and learning: Semiparametric identification and estimation](#). Tech. rep. National Bureau of Economic Research.

References III

-  Butcher, Kristin F, Patrick J McEwan, and Akila Weerapana (2014). “The effects of an anti-grade-inflation policy at Wellesley College”. In: [Journal of Economic Perspectives](#) 28.3, pp. 189–204.
-  Crawford, Gregory S and Matthew Shum (2005). “Uncertainty and learning in pharmaceutical demand”. In: [Econometrica](#) 73.4, pp. 1137–1173.
-  Denning, Jeffrey T et al. (2022). “Why have college completion rates increased?” In: [American Economic Journal: Applied Economics](#) 14.3, pp. 1–29.
-  Denning, Jeffrey T et al. (2025). “Easy A’s, Less Pay: The Long-Term Effects of Grade Inflation”. In.
-  Guvenen, Fatih et al. (2020). “Multidimensional skill mismatch”. In: [American Economic Journal: Macroeconomics](#) 12.1, pp. 210–244.

References IV

-  Hsu, Julian (2018). “Learning about college major match: Microfoundations from dynamic course-taking”. In: [Available at SSRN 3274259](#).
-  Jiang, Xuan et al. (2021). “A second chance at success? Effects of college grade forgiveness policies on student outcomes”. In: [NBER Working Paper w29493](#).
-  Larroucau, Tomás and Ignacio Rios (2020).
[Dynamic college admissions and the determinants of students’ college retention](#).
-  Lovenheim, Michael and Jonathan Smith (2023). “Returns to different postsecondary investments: Institution type, academic programs, and credentials”. In:
[Handbook of the Economics of Education](#). Vol. 6. Elsevier, pp. 187–318.
-  Papageorgiou, Theodore (2014). “Learning your comparative advantages”. In:
[Review of Economic Studies](#) 81.3, pp. 1263–1295.

References V

-  Rojstaczer, Stuart and Christopher Healy (2012). “Where A is ordinary: The evolution of American college and university grading, 1940-2009”. In: [Teachers College Record](#) 114.7, pp. 1–23.
-  Stinebrickner, Todd and Ralph Stinebrickner (2012). “Learning about academic ability and the college dropout decision”. In: [Journal of Labor Economics](#) 30.4, pp. 707–748.
-  Webber, Douglas (2013). “A consolidated patrimonial democracy? Democratization in post-Suharto Indonesia”. In: [Democratization in the Muslim World](#). Routledge, pp. 34–58.
-  Wiswall, Matthew and Basit Zafar (2015). “Determinants of college major choice: Identification using an information experiment”. In: [The Review of Economic Studies](#) 82.2, pp. 791–824.
-  — (2018). “Preference for the workplace, investment in human capital, and gender”. In: [The Quarterly Journal of Economics](#) 133.1, pp. 457–507.

References VI

-  Wiswall, Matthew and Basit Zafar (2021). “Human capital investments and expectations about career and family”. In: [Journal of Political Economy](#) 129.5, pp. 1361–1424.
-  Zafar, Basit (2011). “How do college students form expectations?” In: [Journal of Labor Economics](#) 29.2, pp. 301–348.
-  — (2013). “College major choice and the gender gap”. In: [Journal of Human Resources](#) 48.3, pp. 545–595.